The New Gold Rush: Modelling Bitcoin Volatility with Economic Indicators and Free Sentiment Analysis

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Abstract

This dissertation develops and compares predictive models for Bitcoin prices using freely accessible data, focusing on ARIMA, ARDL, Random Forests, and Linear Regression. The study evaluates both in-sample and out-of-sample performance, incorporating economic indicators and sentiment data. Results show that while Random Forest and Linear Regression models perform well within historical data, they overfit and struggle with future predictions. In contrast, the ARIMA and Scaled ARDL models offer more reliable out-of-sample forecasts. The research highlights the need for balanced model evaluation, particularly in volatile markets like Bitcoin, and suggests hybrid approaches as promising avenues for future development.

Acknowledgments

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Table of Contents

I. Chapter 1: Introduction

Bitcoin, the first cryptocurrency, emerged as a unique financial asset in 2009, later attracting immense interest from both individual and institutional investors. Originally valued around $1 in 2009, by 2024 Bitcoin's value grew to over $73,000. Despite its growing acceptance, Bitcoin remains an exceptionally volatile asset, with price fluctuations that can be drastic and unexpected. This volatility is often attributed to a variety of factors including sentiment and macroeconomic changes.

The approval of Bitcoin Exchange-Traded Funds (ETFs) has added a new dimension to the market dynamics. ETFs are anticipated to bridge the gap between conventional investment products and digital assets, potentially enhancing Bitcoin's liquidity and investor base. One could argue that the introduction of ETFs lends a new level of legitimacy to cryptocurrencies that was previously lacking. This could pave the way for wider recognition of Bitcoin as a viable store of value. Given that Bitcoin has been dubbed 'digital gold' due to its potential to hedge against inflation, it's logical to explore how the correlation between these two assets can be used to predict Bitcoin's price movements (Bhatt, Ghazanfar, and Amirhosseini, 2023).

Traditionally, gold has been seen as a safe haven and a hedge against inflation. Bitcoin, proposed as a modern equivalent, shares characteristics such as a fixed supply and neither are controlled by any central authority, government, or institution. The correlation between gold prices and Bitcoin's value has been a subject of increasing interest, with some researchers utilising the price of movement of gold as an indicator for the price movement of Bitcoin. Bhatt et al. (2023), Choi and Choi (2024), and Kang et al. (2024) are just some of the many researchers that have explored

the relationship between Gold prices and Bitcoin utilising price prediction machine learning techniques with varying degrees of success.

Other researchers use sentiment analysis, primarily drawing from social media platforms like Twitter and Reddit, to predict fluctuations in Bitcoin's price (Frohmann et al., 2023; Kang et al., 2024). These platforms offer a real-time gauge of public opinion, which previous studies have shown to closely align with movements in Bitcoin's market value. However, access to sentiment analysis data proves difficult to attain or comes with a price tag. Nonetheless, such data is argued to be valuable when predicting Bitcoin prices as Bitcoin has historically been viewed as a volatile asset rather than a stable store of value, making it more susceptible to shifts in public sentiment (Kang et al., 2024).

Given these factors, this project aims to create a predictive model that determines Bitcoin prices by integrating only data that can be accessed for free. Such indicators utilised are common economic indicators and free sentiment data provided by augmento.ai.

A. Problem

Bitcoin, the pioneering cryptocurrency, has experienced remarkable growth since its inception in 2009, reaching valuations as high as $73,000 by 2024. However, this growth has been accompanied by extreme volatility, driven by factors such as macroeconomic shifts, market sentiment, and regulatory events, like the introduction of Bitcoin ETFs. These complexities make Bitcoin a challenging asset to predict, yet crucial for investors seeking to navigate its price fluctuations. The core challenge is to develop a predictive framework that effectively handles this complexity while relying solely on freely accessible data sources, as many existing sentiment analysis tools require costly APIs. This research seeks to harness common economic

indicators and freely available sentiment data to build a comparative predictive framework for Bitcoin prices, focusing on how well different models perform both within known data and when forecasting future values. The objective is to maintain robust predictive insights while adhering to the constraints of limited-cost data availability, aligning with the broader aim of accessible financial prediction.

B. Problem Statement

Develop a robust and adaptive predictive framework for Bitcoin prices that integrates econometric and machine learning methods, considering both in-sample and out-of-sample predictions while focusing on the comparative analysis of different models, using only freely accessible data sources.

1. Objective 1

To evaluate and compare the in-sample performance of ARIMA, ARDL, Random Forests, and Linear Regression models for Bitcoin price prediction. The objective is to identify which model best captures patterns and trends within the training dataset while considering issues like overfitting, model complexity, and data representation.

2. Objective 2

To assess and compare the out-of-sample predictive performance of the selected models when forecasting future Bitcoin prices. This objective prioritizes evaluating how well each model generalizes beyond the data it was trained on, focusing on differences in model stability, volatility handling, and overall robustness rather than solely on accuracy.

3. Objective 3

To refine the integration of sentiment analysis into predictive models by optimising the processing of sentiment data from platforms like augmento.ai. This objective seeks to enhance forecasting accuracy by testing different aggregation and scaling techniques for sentiment indicators, integrating this qualitative data into the quantitative analysis to offer a more comprehensive forecasting ability.

C. Achievement

The primary achievement of this dissertation is the development and comparative evaluation of a comprehensive predictive framework for Bitcoin prices that integrates diverse methodologies and leverages freely accessible data sources. The project prioritises the use of data that can be easily accessed by the average person without cost, ensuring its applicability to a broad audience. By employing ARIMA, ARDL, Random Forests, and Linear Regression, this framework capitalises on the strengths of both traditional statistical techniques and modern machine learning approaches.

ARIMA serves as a robust foundation for capturing time-series data trends, ARDL delves into the relationships between Bitcoin prices and key economic indicators, Random Forests accounts for complex, non-linear interactions within the dataset, and Linear Regression provides a straightforward method to model relationships and predict based on historical data. In addition, these models integrate sentiment analysis as a predictive factor, drawing from research that highlights the correlation between sentiment and Bitcoin price movements. This multi-faceted approach not only deepens the understanding of Bitcoin price dynamics within traditional

financial contexts but also bridges the gap between sentiment and digital currency predictions using freely accessible data.

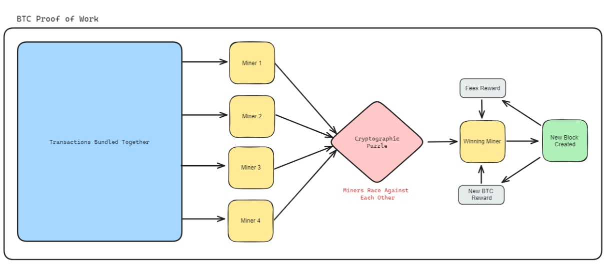
The final outcome is a predictive framework that allows for rigorous comparison of model performance in both in-sample and out-of-sample contexts, offering actionable insights for investors and contributing to academic discussions on cost-effective financial prediction methodologies in the digital age.

II. Chapter 2: Literature Review

A. Bitcoin

Bitcoin, plainly, is a digital currency. When examining Bitcoin's prices, first, one should understand what it means for Bitcoin to be a digital currency and how it operates. Bitcoin utilises blockchain technology, which acts as a decentralised ledger system (Nakamoto, 2009). Proof of Work (PoW) is the consensus mechanism at the heart of Bitcoin's security and operational framework, ensuring not only the reliability of transactions but also the immutability. In the context of Bitcoin's PoW, miners engage in a competition to decipher a cryptographic challenge, and the first one to crack it is awarded the right to append a new block to the blockchain. This reward comes in the form of newly created bitcoins and transaction fees. This process not only fortifies the security of the network but also guarantees that every participant maintains a consistent record of the transaction ledger, since once a block is verified, every node in the network receives an updated copy of the blockchain (Narayanan et al. 2016). An illustration of this process can be seen in figure 1.

Figure 1:



Though Bitcoin's PoW consensus mechanism does have its limitations in terms of absolute privacy and scalability, its mechanisms for transactions and verifications as well as the creation of new coins through mining are proving to be effective in maintaining the reliable and secure digital currency. Moreover, Bitcoin's supply is finite addressing the need for scarcity to ensure a store of value.

Bitcoin transactions occur directly between users without central banks, known as peer-to-peer (P2P); transactions are secured using asymmetric encryption, which involves a public key for encryption and a private key for decryption and signing (Bitcoin.com, 2024). This ensures that only the private key holder can authorise transactions, enhancing security. To make a transfer, a user initiates a transaction from their digital wallet, detailing the sender's and recipient's public addresses, the Bitcoin amount, and a miner's fee. This transaction is then signed cryptographically with the sender's private key, confirming ownership of the bitcoins.

The transaction is broadcast to the Bitcoin network and then it awaits miner confirmation, in the PoW mechanism. Miners then solve the cryptographic puzzles to add the transactions to new

blocks on the blockchain; once a block is verified by the network, it confirms the transactions. For enhanced security, it is advisable to wait for an additional six blocks to be added after the initial confirmation to prevent issues like double spending or transaction reversals during a blockchain fork, thus ensuring the transaction's finality and the system's integrity (Bitcoin.com).

Since its inception, Bitcoin has received both criticisms and praise, as well as significant growth in value. To understand how to go about predicting Bitcoin Prices, it is important to discuss the history of Bitcoin's price volatility, the role of sentiment analysis in predicting Bitcoin prices, other economic indicators that could assist in predicting Bitcoin prices, the potential impact of ETFs on Bitcoin prices, and the various machine learning and statistical methods that could be employed to predict Bitcoin prices.

1. Bitcoin Price Volatility

Bitcoin, launching just a year after the 2008 global financial crisis, captivated global attention due to its groundbreaking technology and potential as an alternative financial system. However, as it was a very new form of technology without the backing of wide acceptance of value, its journey has been marked by significant price volatility. This instability is partly due to its decentralised nature, lack of regulatory oversight, and ongoing debates about its utility and security. It is also impacted by its four year halving cycle in which its daily supply of newly minted Bitcoins is halved.

The price of Bitcoin has experienced dramatic fluctuations, driven by a diverse range of factors including investor sentiment and geopolitical events. This volatility not only makes Bitcoin a fascinating subject for financial analysis but also a challenging asset to price predict due to its volatile nature (Reiff, 2024; Coinage, 2024).

2. Role of Sentiment Analysis in Predicting Bitcoin Price

Due to Bitcoin's high volatility, traditional financial prediction models often fall short; sentiment analysis, however, has emerged as a powerful tool in predicting Bitcoin's price movements. Several independent studies have shown that adding other indicators in addition to price to a forecasting model will aid in the forecast's accuracy and predictions (Critien et al., 2022; Mudassir et al., 2020; Pano & Kashef, 2020). Public sentiment is one such utilised indicator of prices.

Studies have shown that people often make decisions based on emotions rather than logic, and this emotional decision-making can be captured by sentiment analysis of social media data (Gurdgiev & O'Loughlin, 2020). By analysing the mood and opinions expressed in social media, news headlines, and financial reports, researchers have been able to gauge public sentiment towards Bitcoin and utilise these metrics in price prediction models (Dimitriadou & Gregoriou, 2023; Frohmann et al., 2023). Social media sources, like Twitter and Reddit, are often cited as being useful for collecting sentiment analysis data for predicting the price of Bitcoin (Aharon et al., 2022).

This sentiment has a direct impact on its market price as positive news can lead to rapid price increases, while negative news can cause sudden declines. Wolk (2019) asserts that Bitcoin's price fluctuates depending on people's perception and opinions rather than following institutional regulations. This relationship underscores the influence of public perception on Bitcoin,

reflecting its sensitivity to sentiment changes and highlighting the importance of including this metric when developing a model for Bitcoin price prediction (Kang et al., 2024).

Undeniably, sentiment appears to have a direct impact on the price of Bitcoin. However, other indicators should be considered as well. Being that Bitcoin is seen as a store of value and potential hedge against inflation, there is a possibility that other economic indicators may prove useful for Bitcoin price prediction when paired with sentiment analysis data.

3. Other Economic Indicators useful for predicting Bitcoin Prices

As a unique and new form of currency, new relative to typical stores of value, there has been significant research surrounding Bitcoin's properties and macroeconomic impact. However, when it comes to predicting Bitcoin prices, the research varies. Many researchers utilised technical factors, predominantly Bitcoin's hashrate to determine Bitcoin prices (Bouoiyour & Selmi, 2015; Ciaian et al., 2015; Garcia et al., 2014; Georgoula et al., 2015; Hayes, 2015; Kristoufek, 2015).

Based on their research, it is assumed that the hashrate has a positive impact on Bitcoin prices. However, when examined closer, it is clear that this is a case of reverse causality. Typically, in economics, when you see an increase in supply there will be a decrease in price. Kjaerlandet al. (2018) showed that From early 2017 to early 2018, there was a period of rapid growth in both the price of Bitcoin and the hashrate. They explained this phenomenon as the increase in Bitcoin's price encouraged more mining activity, not the other way around. As this is a perfect example of reverse causality, it would mean that the hashrate is not a good explanatory variable for Bitcoin prices and either an instrumental variable or alternate variables are needed to predict Bitcoin prices.

Ciaian et al. (2015) argues that Bitcoin, like other traditional currencies, adheres to some foundational economic principles, such as supply and demand. For instance, research has shown that Bitcoin prices rise when Bitcoin transaction volume increases (Polasik et al., 2015). Moreover, several researchers argue that currency exchange rates, Equity Market Indices, and commodity prices are all fundamental factors that influence Bitcoin prices (Balcilar et al. 2017; Bouoiyour & Selmi 2016; Ciaian et al., 2015).

Gold, for example, is one commodity that Bitcoin is often compared to and assumed to correlate with Bitcoin prices. Oil is another commodity proposed to correlate with Bitcoin prices. Both of these commodities are scarce resources with a technical finite supply, similar to Bitcoin. Gold and Bitcoin are both thought of as a store of value and a hedge against inflation. Ciaian et al. (2015) and Kristoufek (2015) both found that the prices of Bitcoin are positively influenced by the prices of gold and oil. Conversely, Kjaerlandet al. (2018) proved that Gold prices and Oil prices were both insignificant in their ARDL model. Moreover, they observed that the price volatility of Bitcoin was unlike either Gold or Oil. This finding was corroborated in Bouri et al. (2018), who also found that neither an aggregate commodity index nor gold prices have any impact on Bitcoin prices.

The varied research findings highlight the unique position of Bitcoin in the financial landscape, straddling the lines between currency, commodity, and technological innovation. Influenced by a variety of economic and sentiment-driven factors, the predictability of Bitcoin's value poses a challenge to conventional economic models. Given its frequent comparison to gold, it's appropriate to delve deeper into the investment mechanisms that link these assets, with a particular focus on the emerging sector of Bitcoin ETFs.

4. Potential Impact of ETFs on Bitcoin

Though gold was shown in past research to be insignificant in the prediction of Bitcoin prices, researchers agree that Bitcoin and Gold are both inherently similar commodities acting as a store of value and potential hedge against inflation. Moreover, both gold and Bitcoin, much more recently, have approved Exchange Traded Funds (ETFs).

The introduction of ETFs trading gold led to increased prices and common use of gold as a commodity used to hedge against inflation in traditional financial markets (Vardai, 2024; Strack, 2024). By providing a more accessible avenue for investing in gold, ETFs have enhanced its liquidity. Since the introduction of the first gold ETF, the price of gold has increased by a factor of 6.17 (Basar, 2024). See figure 2 for the impact of the ETFs on the price of gold:

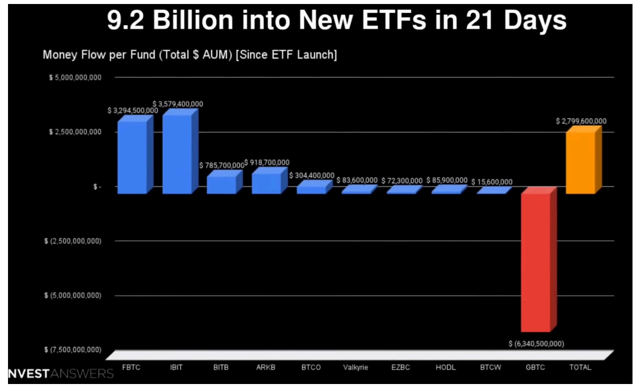
Figure 2: (Altucher, 2024)

Similarly, the recent approval and trading of Bitcoin ETFs are anticipated to have a similar effect on Bitcoin's market (Coinage, 2024; De Mott, 2024; Reiff, 2024). ETFs offer a regulated, investment-friendly avenue, potentially attracting a new wave of institutional and retail investors. In February 2021, Canada launched a spot Bitcoin ETF, Purpose Bitcoin ETF. Similarly, the United States approved eleven Bitcoin spot ETFs on Jan. 10, 2024. Spot Bitcoin ETFs are incorporated in ten countries worldwide and traded in five geographical markets.

The increased demand and enhanced accessibility provided by ETFs is expected to lead to increased prices and wider acceptance of Bitcoin as a legitimate financial asset. Since the introduction of Bitcoin ETFs, significant shifts have occurred in the market. These ETFs are trading large volumes, demonstrating their popularity and highlighting their substantial impact on the market. For example, the Blackrock owned ETF, IBIT, reached $10 billion in assets under management quicker than any other, achieving this milestone in just 37 trading days; only 4% of all ETFs have ever managed to accumulate $10 billion in assets (Basar, 2024).

Conversely, there is a notable decrease in investments in the Grayscale Bitcoin Trust (GBTC) alongside corresponding increases in various ETF funds (Basar, 2024). See figure 3 for a visual representation of this movement.

Figure 3: (Basar, 2024)



The GBTC (Grayscale Bitcoin Trust), once a primary vehicle for institutional and retail investors to gain exposure to Bitcoin without directly handling the cryptocurrency, has seen significant changes following the launch of Bitcoin ETFs. These new ETFs have drastically reduced fees in comparison to GBTC driving market share towards them instead. As shown in figure 4, this shift has been quantified by substantial outflows from the GBTC, which decreased by as much as 50% in its Bitcoin holdings since the launch of spot ETFs (Young, 2023). However, the effect of these ETFs will only be seen in data post 2021, with much of the impact coming in closer to 2024 (Tangem,2024). Prior to 2021, other actors would be needed to predict Bitcoin Prices.

5. *Machine Learning and Statistical Methods for Bitcoin Price Prediction*

The desire to accurately predict the price of Bitcoin is not new. As Bitcoin has grown in value, so has the value of its accurate prediction for investors. Numerous models and indicators have been tested for this purpose, with varying success due to Bitcoin’s significant price volatility.

Notably, Kjaerlandet al. (2018) showed that there is a positive relationship with Bitcoin price and its lag; they utilised an autoregressive distributed lag model (ARDL) to capitalise on this relationship. This shows that the efficient market hypothesis does not hold as to be true past returns should not be correlated with present returns. However, the positive relationship with Bitcoin price and its lag indicate that past returns would correlate with future returns. Given that many investors have limited familiarity with Bitcoin, coupled with the complexity and the rapid influx of new information, quick investment decisions are necessary, so it's plausible to infer that investors are influenced by the momentum of rising prices (Kjaerlandet al., 2018).

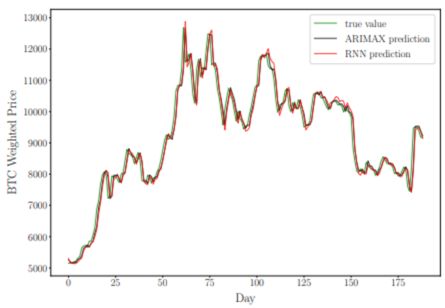
In their research, Kjaerlandet al. (2018) also determined that the hashrate, gold prices, and oil prices are irrelevant when predicting Bitcoin prices. However, they did find that sentiment, like optimism shown through google search volumes, played a significant role in explaining Bitcoin prices. Similarly, Frohmann (2023) utilised Twitter sentiment to predict the price of Bitcoin and employed Long Short-Term Memory models (LTSMs), Temporal Convolutional Networks (TCNs), D-Linear models, and Linear Regression models. The accuracy of these models in Frohmann’s study can be seen in Figure 4.

Figure 4: (Frohmann et al., 2023)

As seen in the results of Frohmann’s (2023) study, the complex models like the LSTM and TCN are unstable and overestimate Bitcoin’s true price, whereas the much simpler models like D-Linear and Linear Regression predict the prices much more accurately showing that these models are better at generalising the given data.

Similar to Frohmann (2023), Serafini et al. (2020) also found that simpler models outperformed more complex models when it came to using sentiment analysis to predict Bitcoin prices. In their case, they employed both an ARIMAX model and an LTSM model to predict Bitcoin prices, discovering that the simpler ARIMAX model surpassed the more intricate LTSM model in performance (Serafini et al., 2020). Their results can be seen in figure 5.

Figure 5: (Serafini et al., 2020)



The insights from these studies underscore a pivotal trend: simpler models often surpass more complex ones in predicting Bitcoin prices, particularly when analysing market sentiment. This trend may reflect the inherent noise and unpredictability in Bitcoin's market dynamics, which can confound more intricate models. The superior performance of simpler models suggests that they may be more adept at filtering out extraneous information and focusing on the most predictive indicators. This finding has significant implications for investors and analysts who seek to harness predictive analytics for investment strategies in the volatile cryptocurrency market. Overall, the continuous refinement and testing of predictive models will likely play a critical role in enhancing the accuracy and reliability of Bitcoin price forecasts.

B. Conclusion

Through past research, the multifaceted dynamics influencing Bitcoin's price volatility can be observed, highlighting the utility of both economic indicators and sentiment analysis in crafting predictive models. Bitcoin's unique position as a digital currency and its parallel trajectory with traditional safe-haven assets like gold are noted by many, including Bhatt et al. (2023), Choi and Choi (2024), and Kang et al. (2024), who have explored the relationship between gold prices and Bitcoin, utilising price prediction machine learning techniques with varying degrees of success. Notably, the research underscored the potential of emerging Bitcoin ETFs to reshape market perceptions and enhance liquidity, thereby further legitimising Bitcoin as a mainstream financial instrument.

A key insight from the literature is the superior performance of simpler predictive models in forecasting Bitcoin prices compared to more complex algorithms, particularly in environments with high volatility and rapidly changing investor sentiment (Frohmann, 2023; Serafini et al., 2020). This finding challenges the notion that sophisticated models always yield better results, suggesting instead that simpler models, which excel at filtering critical information from noise, may be more effective in such contexts.

Moreover, the integration of freely accessible sentiment data and traditional economic indicators offers a promising approach for building cost-effective and accessible predictive models. This aligns well with Bitcoin's decentralised ethos and democratises financial forecasting by reducing reliance on expensive data sources.

As the landscape of digital currencies evolves, so must the predictive methods and models applied to them. The review highlights the importance of adapting forecasting models to the

unique characteristics of Bitcoin, emphasising the effectiveness of simpler, more interpretable models over their complex counterparts. This sets the stage for a more nuanced investigation into model performance, focusing on comparative analysis rather than purely on predictive accuracy.

The overarching goal of this research is to develop a predictive framework for Bitcoin prices that integrates econometric and machine learning methods, using freely accessible data sources. The research aims to evaluate and compare the in-sample and out-of-sample performance of ARIMA, ARDL, Random Forests, and Linear Regression models, with a focus on understanding model robustness and stability rather than solely on accuracy. Additionally, the study will explore the impact of key economic indicators and optimise the integration of sentiment analysis to enhance predictive performance.

The next chapter will outline the methodology used in this study, detailing the approaches, data sources, and techniques for implementing the chosen models. The chapter will explain how each model is calibrated and tested, both within the dataset and for future forecasts, providing a comprehensive framework that combines quantitative economic indicators and qualitative sentiment analysis. This approach not only advances academic research but also offers practical insights for investors and analysts in navigating the volatile world of cryptocurrency markets.

III. Chapter 3: Methodology

A. Introduction

Since its inception in 2009, Bitcoin has emerged as a groundbreaking financial asset, initially valued at around $1 and soaring to over $73,000 by 2024. This remarkable growth has not come without challenges; Bitcoin remains an exceptionally volatile asset, with price fluctuations influenced by a myriad of factors, including macroeconomic changes, market sentiment, and the

evolving regulatory landscape. The introduction of Bitcoin Exchange-Traded Funds (ETFs) has further complicated the market dynamics by bridging traditional investment products with digital assets, enhancing Bitcoin's liquidity and expanding its investor base.

This research aims to address the challenge of predicting Bitcoin prices in this complex environment, particularly considering the volatility introduced by both external economic factors and internal market sentiments. The methodology outlined in this chapter focuses on developing a predictive framework using only freely accessible data sources. The goal is to provide reliable predictions without incurring high costs typically associated with advanced data analytics tools.

The objectives of this research are threefold:

1. Objective 1: Evaluate and Compare In-Sample Predictive Models

   This objective focuses on determining which predictive model—ARIMA, ARDL, Random Forests, or Linear Regression—best captures patterns within the training dataset. Emphasis is placed on understanding each model's ability to handle both linear and non-linear aspects of Bitcoin's price dynamics while assessing potential issues such as overfitting, model complexity, and data representation.

2. Objective 2: Assess and Compare Out-of-Sample Predictions

   This objective prioritises evaluating how well each model generalises when predicting future Bitcoin prices beyond the original dataset. The focus is on comparing model stability, handling of volatility, and overall robustness, with less emphasis on pure accuracy metrics. The analysis aims to uncover differences in how each model performs when faced with unseen data.

3. Objective 3: Integrate and Optimise Sentiment Analysis

This objective seeks to refine how sentiment analysis is incorporated into predictive models by testing various techniques for processing sentiment data from freely available platforms like augmento.ai. By exploring different methods for aggregating and scaling sentiment indicators, the goal is to enhance the models' forecasting capabilities and better quantify the impact of public sentiment on Bitcoin prices.

The methodology section will provide a detailed overview of the theoretical frameworks underlying each predictive model and describe the specific techniques used to integrate various data sources. Additionally, the chapter will cover the data collection processes, criteria for selecting data sources, and methods for analysing data to ensure robustness and reliability. By combining traditional statistical approaches with modern machine learning techniques, this research aims to build a predictive framework that is not only effective but also cost-efficient, aligning with the decentralised and open-access ethos of Bitcoin.

The resulting predictive model is expected to deliver insights that support better decision-making for investors while advancing the academic understanding of Bitcoin's price dynamics. The methodological approach seeks to bridge the gap between sentiment and digital currencies using freely accessible data, setting a precedent for cost-effective financial analysis.

B. Approach

This research combines quantitative econometric models and machine learning techniques to evaluate the effectiveness of prediction models for the price of Bitcoin. This strategic methodology utilises and analyses the interplay between economic indicators, sentiment data,

and of course, Bitcoin prices within Linear regression, Random Forests, ARDL, and ARIMA prediction methods.

The Machine Learning models utilised include Linear Regression and Random Forests. As a baseline model, linear regression will be used to estimate the direct impact of economic indicators and sentiment on Bitcoin prices. Random Forests ensemble learning method will be employed to capture nonlinear relationships and interactions between various predictors and Bitcoin prices.

The statistical models and methods utilised include ARDL (Autoregressive Distributed Lag) and ARIMA (Autoregressive Integrated Moving Average). ARDL will help analyse the relationships between Bitcoin prices and lagged values of predictor variables, including economic indicators and sentiment indices. ARIMA will be used to capture the time-series aspects of Bitcoin prices, considering its historical price volatility.

According to the background research showcased in Chapter One, simpler models, like Linear Regression and Random Forests, tend to more accurately predict Bitcoin prices (Frohmann, 2023); Serafini et al., 2020). However, research also shows that Bitcoin prices are correlated with their lag, suggesting that ARDL and ARIMA may both be suitable models for predicting Bitcoin prices as well (Kjaerlandet al., 2018). This study will utilise all mentioned models to predict Bitcoin prices and determine their comparative performance based on economic indicators and sentiment data.

Prior to this analysis the chosen data will undergo preprocessing to handle any missing values, normalise scales and dates, and amalgamate any necessary columns. Feature importance from Random Forests will be utilised to identify the most predictive features and various data sets will

be created to explore which features do, in fact, create the best predictive model. Sentiment data will be systematically incorporated into the predictive models, sourced from augmento.ai, which offers free, pre-scored sentiment data. This data will undergo a rigorous feature selection process to identify the most impactful variables. This data will be amalgamated based on logical assessment and empirical testing to ensure that the integration of sentiment data optimises the predictive accuracy of the models.

Model performance will be evaluated using metrics such as RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), and MSE (Mean Squared Error) to compare the predictive accuracy of different models. The different data sets each containing different amounts of features will also be compared and evaluated. Through this methodology, predictive models will be developed, which model, ARIMA, ARDL, Random Forests, or Linear Regression, has the best predictive accuracy will be determined, and the predictive relationship between economic indicators, sentiment data and Bitcoin prices will be explored.

C. Data Sources

Economic indicators and Sentiment Data in relation to Bitcoin are all utilised for the predictive models. The primary dataset contains historical data on Bitcoin prices (BTC-USD), the S&P 500 index (^GSPC), and major foreign exchange rates (EUR/USD, GBP/USD, JPY/USD) collected from Yahoo Finance for the period from November 1, 2016, to May 22, 2024. This specific time frame corresponds to the availability of free sentiment data. Additionally, economic indicators such as the 10-year Treasury yield and the Consumer Price Index (CPI) were obtained from the Federal Reserve Economic Data (FRED) for the same period. These elements are critical as they potentially influence Bitcoin prices, based on findings from prior research.

1. Analysis and Cleaning

a) Sentiment Data

Free sentiment data was downloaded from Augmento.ai. This data included hourly sentiment scores for Bitcoin for the time period of November 1, 2016, to May 22, 2024. The dataset included a wide array of sentiment-based columns sourced from platforms such as Twitter, Reddit, and Bitcointalk, reflecting various emotional and informational states relevant to Bitcoin and the broader cryptocurrency market. These include positive sentiment indicators like twitter_good_news and reddit_optimistic, which capture optimistic and hopeful messages, and twitter_euphoric_excited and bitcointalk_bullish, highlighting periods of heightened enthusiasm and bullish market outlooks. Negative sentiment indicators such as twitter_bad_news and reddit_pessimistic_doubtful track adverse news and sceptical views, while twitter_angry and reddit_sad reflect anger and sadness in community reactions. Additional specialised columns like twitter_airdrop and reddit_positive cover specific events or general positive sentiments. This diverse range of columns allows for a nuanced analysis of how different sentiments correlate with Bitcoin price movements.

However, since this data was hourly and the economic indicators were daily, the sentiment data needed to be aggregated to achieve daily scores. An aggregation dictionary was created to specify how each column in the hourly data should be summarised when aggregating to daily data. For columns containing sentiment scores, the method chosen was 'sum', meaning that all hourly scores for a given day were summed up to provide a total daily sentiment score. This approach captured the cumulative sentiment expressed throughout the day. For price-related data, such as the 'listing_close' column, the aggregation method specified was 'last'. This method ensured that the closing price of the asset for each day is captured by taking the last available price in the hourly data for that day. The resample('D') function was used to change the frequency of the data from hourly to daily. The .agg(aggregation_methods) function then applied the specified aggregation methods to each column, resulting in a DataFrame with daily data. I now had a daily sentiment data frame called daily_data.

b) Economic Indicators

Using Python's reduce function, the intersection of dates from all data sources (Bitcoin, S&P 500, and forex rates) was computed, creating a common index. This step was crucial to ensure that all data points correspond to the correct dates for accurate comparative analysis. Each dataset was reindexed to this common index, standardising the dates across datasets which helps in maintaining consistent analysis periods across different types of data. All individual datasets were then combined into a single DataFrame. Missing values were handled by forward and backward filling to ensure no gaps in the data, preserving the integrity for time series analysis. I now had a data frame called data that contained all necessary economic indicators.

c) Combined Data Sets

Using daily_data (Sentiment) and data (Economic indicators), I was able to merge these data sets into one data frame called data_all which contained 287 columns, six of which were economic indicators, one of which was the price of Bitcoin, and the rest were sentiment related. The sentiment-related column contained daily sentiment scores for various topics pertaining to Bitcoin on Reddit, Twitter, and Bitcointalk, see figure 6.

Figure 6.

```
print(data_all.columns.tolist())
```

```
['Date', 'S&P500', 'EUR_USD', 'GBP_USD', 'JPY_USD', '10Y_Treasury_Yield', 'CPI', 'BTC_close', 'twitter_hacks', 'twitter_pess
imistic_doubtful', 'twitter_banks', 'twitter_selling', 'twitter_market_manipulation', 'twitter_de_centralisation', 'twitter_
angry', 'twitter_etf', 'twitter_leverage', 'twitter_bottom', 'twitter_institutional_money', 'twitter_fomo', 'twitter_predict
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'reddit_rumor', 'reddit_scam_fraud', 'reddit_airdrop', 'reddit_optimistic', 'reddit_negative']
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Of course, this was a lot of information that may or may not have an impact on the price of Bitcoin. Feature selection was performed to ascertain the most import features for Bitcoin price prediction.

The feature selection process was undertaken with a focus on refining the predictive variables for forecasting Bitcoin prices. Initially, correlation analysis was conducted to identify features that

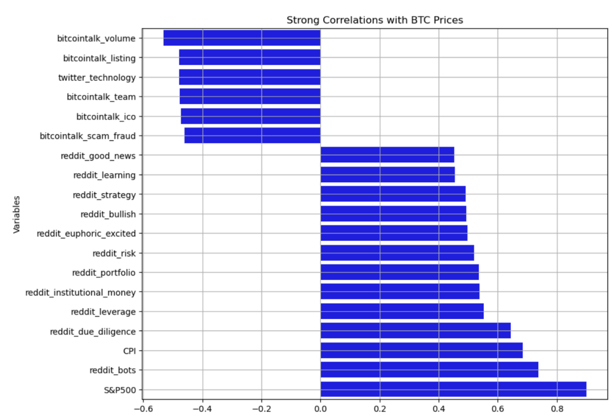
exhibited a significant linear relationship with the Bitcoin closing prices (BTC_close). Features that had an absolute correlation coefficient greater than 0.3 were considered significant and included in a preliminary list for further analysis. Variables with correlations greater than 0.3 were chosen in order to include multiple variables with both strong and low to moderate levels of correlation (Mukaka, 2012).

To refine this list and address multicollinearity among the selected features, the Variance Inflation Factor (VIF) was employed. Multicollinearity can significantly distort the reliability of a linear model; hence, the VIF method helps identify and exclude highly correlated independent variables (Kim, 2019). All features with a VIF greater than 10 were considered to have high multicollinearity and were excluded from the model to ensure the robustness of the predictive analysis.

This methodological approach led to a list of features where the correlations were sufficiently strong and independent of each other to justify inclusion in the final model. The features retained included: 'bitcointalk_launch', 'bitcointalk_bots', 'bitcointalk_angry', 'bitcointalk_roadmap', 'bitcointalk_announcements', 'bitcointalk_sad', 'bitcointalk_partnerships', 'twitter_scaling', 'bitcointalk_bug', 'bitcointalk_going_short', 'bitcointalk_whitepaper', 'reddit_etf', 'twitter_volume', 'reddit_stablecoin', 'reddit_tax', 'reddit_leverage'.

Two economic factors, S&P500 and CPI, were also included in the selected features despite having high VIFs, in order to ensure that the data set included both sentiment data and economic indicators. Those two economic indicators were chosen due to their strong correlations, see figure 7. These features were deemed critical as they could provide nuanced insights into the factors driving Bitcoin price movements.

Figure 7.



Through feature selection and logical application, a new data set was created with the selected features, the chosen economic indicators, and Bitcoin Price, called filtered_data.

A final data set, called ARDL_data was then created to amalgamate similar sentiment data to mitigate issues of cointegration when running ARDL models. In time series analysis, especially when dealing with economic and sentiment data that are non-stationary, cointegration can lead to misleading results about the influence of independent variables on the dependent variable. Cointegration implies that one or more combinations of these variables can have a long-term equilibrium relationship, which complicates individual effect analysis. The amalgamation of similar sentiment data into a final dataset for ARDL (Autoregressive Distributed Lag) models serves a critical role in ensuring the robustness of the econometric analysis.

By grouping similar types of sentiment data into broader categories like 'Positive', 'Negative', 'Product_Technology', 'Investment', 'Community', and 'Regulation', the dataset becomes more manageable, and it addresses potential multicollinearity and cointegration among highly correlated variables. This categorization not only reduces the number of predictors in the ARDL model but also focuses on the overall impact of these broader sentiment categories on Bitcoin prices. The process involved mapping each sentiment-related column to one of the predefined categories and then calculating the sum of all columns within each category for each observation.

This aggregation transforms the granular sentiment data into composite scores that represent the collective sentiment under each category. The ARDL_data data set contained the following: S&P500, EUR_USD, GBP_USD, JPY_USD, 10Y_Treasury_Yield, CPI, BTC_close, Positive, Negative, Product_Technology, Investment, Community, Regulation.

With three distinct datasets—data_all, filtered_data, and ARDL_data—at hand, we are well-equipped to execute various models and derive insights into how different features influence Bitcoin prices. This comprehensive approach ensures a thorough exploration of both economic and sentiment drivers, enhancing our understanding of cryptocurrency market dynamics.

2. Machine learning and Statistical Methods

A combination of machine learning and statistical models is employed to capture the dynamic and multifaceted nature of Bitcoin price prediction. Specifically, Linear Regression, Random Forests, ARDL (Autoregressive Distributed Lag), and ARIMA (Autoregressive Integrated Moving Average) models are utilised, each chosen for their unique capabilities and suitability to the structured datasets—data_all, filtered_data, and ARDL_data.

The data_all, filtered_data, and ARDL_data datasets were systematically utilised across different models to optimise the analysis of Bitcoin price influences. Both Linear Regression and Random Forests were applied to each of the three datasets—data_all, filtered_data, and ARDL_data. This comprehensive application helps to validate the robustness of the findings across different types of machine learning techniques and data structures.

The ARIMA and ARDL models, however, were exclusively utilised with the ARDL_data dataset. The rationale behind this specific pairing lies in the ARDL model's ability to address varying levels of integration among variables, making it particularly suited for the ARDL_data dataset, which had been refined to address issues of multicollinearity and cointegration. This dataset, structured with aggregated sentiment categories and streamlined economic indicators, provides an ideal basis for exploiting the ARDL model's capabilities in analysing both short-term dynamics and long-term equilibrium relationships. Similarly, the ARIMA model's proficiency in handling non-stationary data makes it a good fit for the ARDL_data, where its application is geared towards exploiting historical price data and trend-based features inherent in Bitcoin's price movements.

By tailoring the use of each model to datasets prepared to highlight their strengths, this research ensures a nuanced approach to the predictive analysis of Bitcoin prices, leveraging the specific advantages of each statistical and machine learning methodology. This targeted application of models to datasets is designed to maximise the interpretability and reliability of the results, providing a robust analytical foundation for understanding the factors influencing Bitcoin prices.

D. Evaluation

The predictive models were evaluated using a range of performance metrics, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2). These metrics were selected for their ability to assess different aspects of model accuracy and reliability, thereby providing a comprehensive perspective on each model's predictive capability.

Mean Squared Error (MSE) measures the average squared difference between the actual and predicted values, heavily penalising larger errors. This makes it particularly useful for models where large deviations are especially undesirable. Root Mean Squared Error (RMSE), derived from MSE, shares the same units as the target variable—Bitcoin prices—offering more interpretable insights into the magnitude of prediction errors. This metric is particularly valuable for comparing models with different data scales. Mean Absolute Error (MAE) calculates the average absolute difference between predicted and actual values, making it a robust measure that is less sensitive to outliers compared to MSE and RMSE. R-squared (R^2) represents the proportion of the variance in Bitcoin prices that can be explained by the independent variables, with higher values indicating better model performance. However, this metric must be interpreted cautiously in time-series analysis to avoid overfitting (Hodson, 2022).

After determining the best-performing models within each category—Linear Regression, Random Forests, ARIMA, and ARDL—the selected models were then utilised to forecast Bitcoin prices for the next 30 days beyond the historical data period. This out-of-sample forecasting allowed for a realistic assessment of how well each model generalises to unseen data, which is critical for real-world applications. The same evaluation metrics were applied to these forecasts to maintain consistency and comparability.

For each model type, the best-performing model was chosen based on a combination of low error metrics (MSE, RMSE, MAE) and high R^2 values within the given datasets. These models were then tasked with predicting Bitcoin prices for the 30 days outside of the dataset. The forecasts were subsequently compared using the same metrics to determine how well each model maintained its predictive accuracy in an out-of-sample context.

This comparative analysis across both in-sample and out-of-sample forecasts provided a robust evaluation framework. It highlighted which models were most effective at capturing the complex dynamics between Bitcoin prices, economic indicators, and sentiment data, and which models showed the strongest potential for reliable future predictions. This methodology ensured that the final selection of models was not only based on their performance within the historical dataset but also on their ability to anticipate price movements in new data.

Several challenges emerged during model evaluation. Multicollinearity was a significant issue, particularly for Linear Regression models, ARDL models, and ARIMA methods, which are sensitive to correlated predictors. To mitigate this, the Variance Inflation Factor (VIF) was used to filter out highly correlated features, resulting in more stable and reliable Linear regression and Random Forest models.

The sentiment data, collected from multiple platforms, introduced inherent noise and variability into the models. Aggregating this data into broader categories for the ARDL model and ARIMA estimation proved essential in reducing noise and enhancing model stability. This aggregation allowed for more focused analysis by grouping related sentiment data into comprehensive categories such as "Positive," "Negative," "Product_Technology," "Investment," "Community," and "Regulation". By balancing model complexity with interpretability and accuracy, and by employing a diverse set of metrics, the evaluation provided a comprehensive understanding of model performance.

IV. Chapter 4: Development and Testing

A. Linear Regression

Four Linear Regression Models were run, each using a different Data Set. The first, Linear regression Model 1, utilised the filtered_data data set. This data set had its features selected in the feature selection process discussed previously, as well as CPI and S&P 500 ('bitcointalk_launch', 'bitcointalk_bots', 'bitcointalk_angry', 'bitcointalk_roadmap', 'bitcointalk_announcements', 'bitcointalk_sad', 'bitcointalk_partnerships', 'twitter_scaling', 'bitcointalk_bug', 'bitcointalk_going_short', 'bitcointalk_whitepaper', 'reddit_etf', 'twitter_volume', 'reddit_stablecoin', 'reddit_tax', 'reddit_leverage').

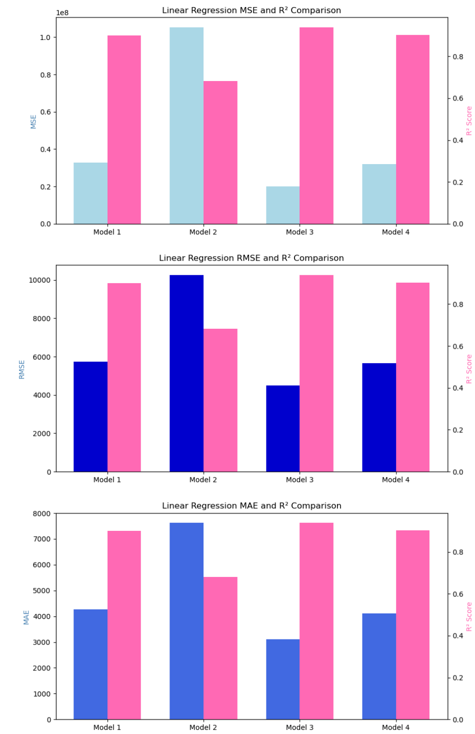
The second, Linear regression Model 2, also utilised the filtered_data set; however, this time it only included the features chosen in the feature selection and left out the CPI and S&P 500.

The third, Linear regression Model 3, utilised the data_all data set which included all features with none taken out.

The fourth, Linear regression Model 4, utilised the ARDL_data data set which had the amalgamation and mapping of the sentiment data as well as all of the economic indicators ('S&P500', 'EUR_USD', 'GBP_USD', 'JPY_USD', '10Y_Treasury_Yield', 'CPI', 'BTC_close', 'Positive', 'Negative', 'Product_Technology', 'Investment', 'Community', 'Regulation').

All four models were compared using MSE, RMSE, MAE and their R-squared. See figure 8.

Figure 8.

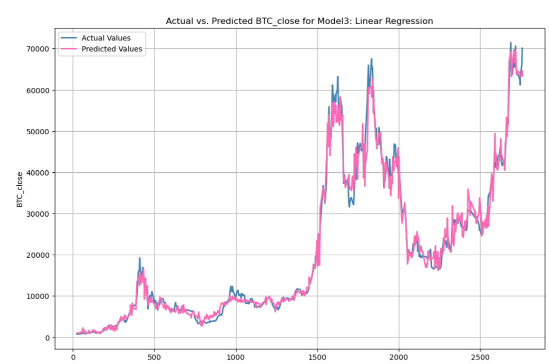


Based on the analysis presented in Figure 8, Linear Regression Model 3 is notably the most effective for predicting Bitcoin prices, as evidenced by its low error metrics and a high R^2 value of 0.939. This model utilised the complete data_all dataset, which retained all original features, suggesting that the inclusion of diverse features captured valuable interactions crucial for accurate Bitcoin price forecasting. See the performance of Linear Regression Model 3 when predicting Bitcoin prices within the data set.

Models 1 and 4, despite using different datasets, both achieved R-squared values around 0.900. These results indicate substantial predictive capabilities, although they did not reach the optimal levels observed in Model 3. Model 1 used the filtered_data dataset with additional economic indicators, while Model 4 used the ARDL_data dataset, which incorporated categorised sentiment data alongside economic indicators.

Model 2, which also utilised the filtered_data set but excluded economic indicators, exhibited the highest error metrics and the lowest R^2 value at 0.681. This performance suggests that excluding economic indicators significantly compromises the model's effectiveness.

Figure 9.



Although Model 3 shows superior performance, the generally high R-squared values across the models raise concerns about potential overfitting. The highly accurate fitting can be seen further in Figure 9. In the context of Bitcoin price prediction, an R^2 value between 0.5 and 0.9 might be more indicative of a balance between adequate fit and generalisation, particularly given the complexity of financial markets. The high R^2 values observed might imply that the models are capturing noise rather than underlying patterns, especially since the dataset that performed best was the unfiltered data_all.

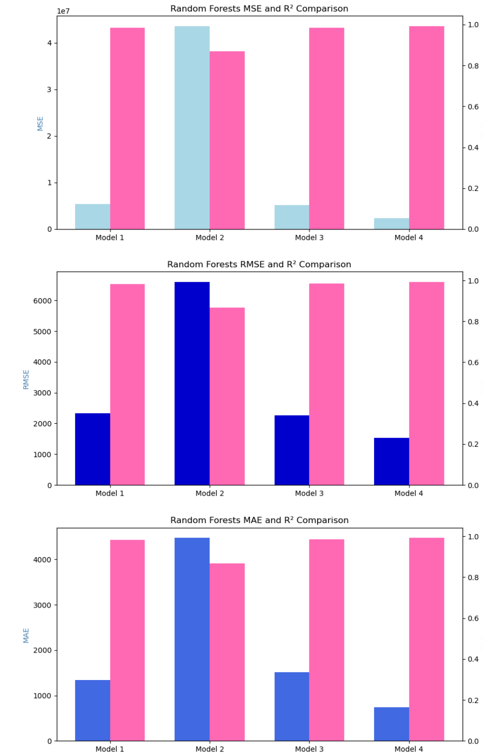
B. Random Forests

Similarly to the Linear Regression models, four Random Forests models were also executed, each utilising a distinct dataset. Random Forest Model 1 applied the filtered_data dataset, which includes features selected during the feature selection process as well as key economic indicators such as CPI and S&P 500, all of which are listed in the Linear Regression section. Random Forest Model 2 utilised the same filtered_data dataset but excluded the economic indicators, focusing solely on the features identified in the feature selection. Random Forest Model 3 employed the data_all dataset, which includes all available features without any exclusions, providing a comprehensive view of potential predictive variables. Random Forest Model 4 used the ARDL_data dataset, which consolidated sentiment data into broader categories alongside all economic indicators. For consistency, all models were evaluated using metrics such as MSE, RMSE, MAE, and their R-squared to gauge their predictive effectiveness (see Figure 10).

Analysis of the results show that Random Forest Model 4 was the most effective, demonstrating superior predictive accuracy with the lowest error rates across all metrics and the highest R^2 value of 0.993. However, there is obvious overfitting given the extremely high R-Squared values in each Random Forests Models. Regardless, for the given data sets, Model 4's low errors suggests that the combination of categorised sentiment data and comprehensive economic indicators significantly enhances the model's ability to forecast Bitcoin prices accurately within the data set. The overfitting and high accuracy of predictions for Model 4 can be seen further in Figure 11.

Random Forest Model 3 also showed excellent predictive ability, albeit with a very high R^2 value of 0.984, however this does further underscoring the advantage of utilising a complete dataset that retains all features for predictions within the given data set.

Figure 10.



In contrast, Model 1, though employing a filtered set of predictors including economic indicators, still performed well with an R^2 value of 0.984 but did not achieve the error metrics as low as Model 4. This aligns with the understanding that while selective features can provide strong predictions, the aggregation and comprehensive analysis offered by Model 4 offer a more nuanced understanding.

Model 2, which excluded economic indicators, showed significantly weaker performance with the highest error rates and a lower R^2 value of 0.868. This result highlights the critical role of economic indicators in enriching the model's dataset, leading to better predictive performance.

Figure 11.

The high R^2 values observed across the models, particularly for Models 3 and 4, suggest a strong fit, but most likely overfitting, to the historical data, affirming the models' capabilities in capturing the intricacies of Bitcoin price movements within the given data set as seen in Figure 11. However, the overfitting for both the best performing Linear Regression Model and Random Forests model may suggest that while the models perform well within the data, forecasting future values may have a significantly lower accuracy.

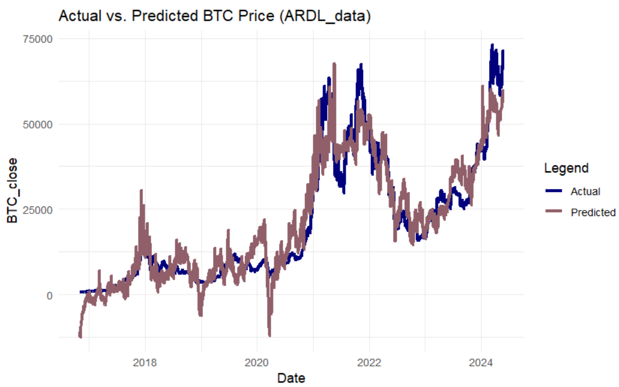
C. ARDL

When attempting to run an ARDL model, there was a significant issue of multicollinearity in the data_all data set. This issue was removed through creation of the ARDL_data data set which amalgamated the sentiment data into various categories. The ARDL_model was then run, and the following metrics were derived: MSE of 32,575,592, RMSE of 5707.503, MAE of 4334.558, and an R^2 of 0.9011.

The high MSE and RMSE values indicate substantial average squared errors and average errors per prediction, respectively. This suggests that while the model captures a lot of the data's variability, its predictions can deviate significantly from actual prices. The MAE being 4334.558 supports this interpretation, showing that typical predictions can be off by over $4,000 from actual prices, which is considerable but perhaps expected given the volatile nature of Bitcoin.

Once again, the model has a rather high R-Squared score. The R-Squared of 0.9011 is quite high, suggesting that the model explains about 90% of the variance in Bitcoin prices from the predictors used; while impressive, this could also hint at potential overfitting, especially given the complex nature of the predictors involved. See figure 12 to examine the ARDL model's accuracy of predictions within the data.

Figure 12.



Both the actual and predicted lines follow similar trends, particularly in capturing major movements in the price of Bitcoin. This includes the significant rises and subsequent declines observed around 2021 and 2023. There is a close match between the predicted and actual prices for most of the timeline, indicating that the ARDL model is relatively effective in understanding the factors that influence Bitcoin prices. However, during periods of extreme price spikes or drops, the model's predictions diverge somewhat from actual prices, suggesting limits to its predictive accuracy under highly volatile market conditions.

Similar to the Linear Regression and Random Forest models, while the ARDL model tracks the historical data quite closely, this could also be indicative of overfitting, where the model is possibly too finely tuned to the historical data, which may affect its performance on new, unseen

data. See Figure 13 for a comparison of the best Linear Regression Model, ARDL Model, and Random Forests Model.

Figure 13.

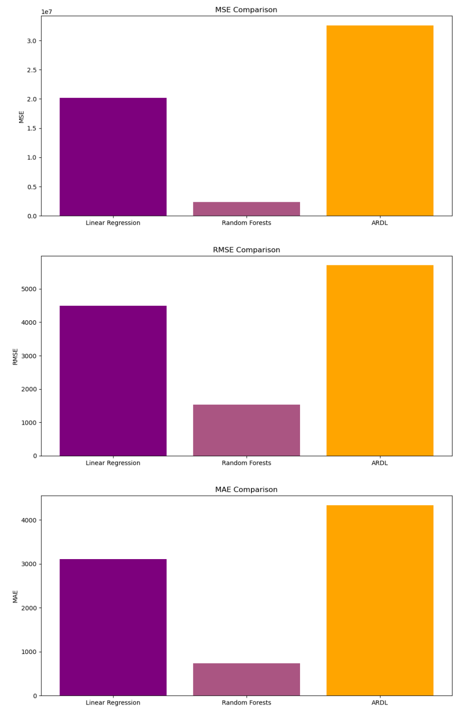


Figure 13 highlights that the Random Forests model was the most accurate in predicting Bitcoin prices within the given dataset. This aligns with the background research which indicates that simpler models tend to better predict Bitcoin Prices. Random Forests most likely excelled due to its ability to capture complex, non-linear relationships, making it highly effective for in-sample predictions. However, this also increases the risk of overfitting. To best explore the model's predictive accuracy, future predictions outside of the data need to be made.

D. Forecasting Future Values

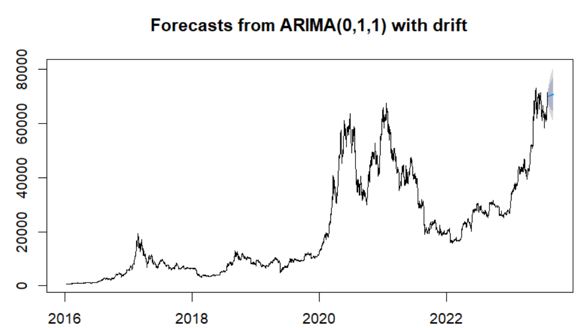
1. ARIMA

The ARDL_data data set was also fitted using the ARIMA estimation method. For this, the following metrics were observed: MSE: 930742.7, RMSE: 964.7501, MAE: 519.7464, R-Squared: 0.9971746. ARIMA (Autoregressive Integrated Moving Average) differs from ARDL (Autoregressive Distributed Lag), Linear Regression, and Random Forests primarily in its approach and application, particularly in handling time series data. In this case, the original ARDL_data data was not stationary and required first-differencing to achieve stationarity. The successful transformation into a stationary series was verified using the Augmented Dickey-Fuller (ADF) test, confirming that the differenced data did not contain trends or seasonal effects that could bias the forecast.

The performance metrics resulting from this model were exceptionally high, with an R-squared value of 0.9971746, indicating that the model could explain nearly all the variance in the Bitcoin price movements within the dataset. This high degree of accuracy, reflected also by a fairly low MSE (930742.7) and RMSE (964.7501) scores, and a minimal MAE (519.7464), underscores the

model's capability in capturing the underlying patterns of the data. Once again, however, there is potential for overfitting. See figure 14 for the ARIMA forecast.

Figure 14.



Given that the Linear Regression model, Random Forests model, ARDL model, and ARIMA estimations all seem to have overfitting but predict within the data set accurately, the next step was to see how they might perform at predicting values outside of the data set, ergo forecasting the next 30 days for the Linear Regression, Random Forests, and ARDL models. For this, the best models of each type were utilised.

2. Linear Regression

The best-performing Linear Regression model was used to predict Bitcoin prices for the 30 days following the last date in the dataset, from May 23, 2024, to June 21, 2024. The prediction process was iterative and involved feeding back the predicted prices into the model to predict the next day's price. The model was trained using 75% of the available data (data_all) and tested on the remaining 25%. This ensures that the model is robust and generalises well on unseen data.

The future predictions were made iteratively. The predicted price of one day was used as input for predicting the next day's price. The model was initially fed the last known values from the training dataset, including features such as the S&P 500, CPI, and others. Additionally, the economic indicators were adjusted slightly to simulate small growth or inflation over time (e.g., a 0.1% increase in the S&P 500 and 0.05% increase in CPI daily). After predicting the future prices, a scaling factor was applied to align the first predicted value with the actual last observed Bitcoin price. This adjustment was necessary to reduce any potential bias that might arise due to slight differences between the model's initial prediction and the actual data.

The predicted values suggest a steady upward trend in Bitcoin prices over the 30-day forecast period. The consistent growth aligns with the model's understanding of the historical data, which likely captured long-term bullish sentiment and macroeconomic factors influencing Bitcoin prices. However, it's important to note that linear regression models tend to smooth out fluctuations and might not fully capture sudden market shifts or volatility often seen in cryptocurrency markets. This smoothing out resulted in a seemingly straight line prediction, as seen in Figure 15.

Figure 15.

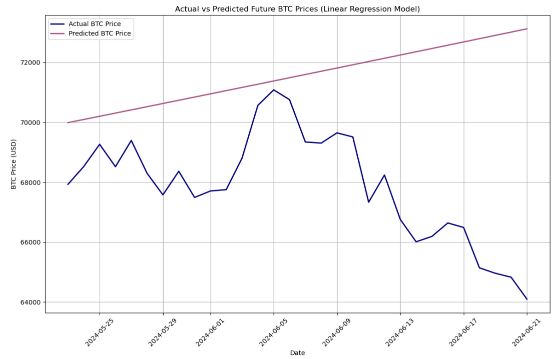


Figure 15 distinctly shows that the Linear Regression model consistently predicted higher Bitcoin prices across the forecasted period, showing a steady, linear increase. This model forecasted prices in the range of $69,000 to $73,000, indicating a strong upward trend. However, these predictions deviate significantly from the actual observed prices. The smooth and overly optimistic trend suggests that the Linear Regression model fails to capture short-term fluctuations and market volatility. This is likely due to the model's inherent assumption of linear relationships and the high sensitivity to outliers, leading to overfitting. Consequently, while this model might be useful in stable markets, it underperforms in highly volatile environments like the Bitcoin market.

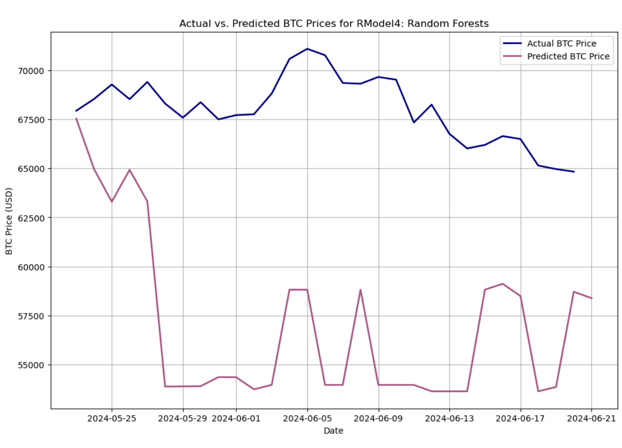
3. Random Forests

Similarly, the best-performing Random Forest model was used to predict Bitcoin prices for the same 30-day period, from May 23, 2024, to June 21, 2024. This process involved generating

synthetic future feature values using a random distribution and predicting future Bitcoin prices based on these simulated features. Unlike the linear regression approach, where the previous day's predictions were iteratively fed into the model, the Random Forest method utilised a different approach. Random changes (with a normal distribution) were applied to each feature to simulate how those features might behave in the future.

The Random Forest model then took these synthetic future features and predicted Bitcoin prices for each of the 30 days. The model's strength lies in its ability to handle complex, nonlinear relationships and interactions between features, which might lead to better results than Linear Regression given Bitcoin's volatility. See Figure 16 for the results.

Figure 16.



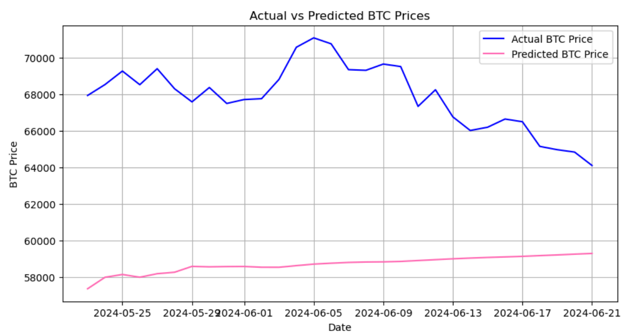
Based on Figure 16, the Random Forest model's predictions do appear to have more fluctuations than that of the Linear Regression Model. The model begins by predicting prices close to the actual values but quickly diverges as it progresses. Random Forests tend to perform well when capturing complex, non-linear interactions among features, but in this case, the model produced inconsistent predictions, likely due to overfitting during training. The erratic jumps in the forecasted values are a clear indication that the model struggles to generalise when predicting future unseen data, highlighting its limitations in time-series forecasting where smooth trends are expected.

4. ARDL

The ARDL model's future prediction process was more complex than the simpler linear regression or random forest models. It involved generating forecasts for each of the predictors independently, followed by calculating the predicted Bitcoin closing prices based on those forecasts using the ARDL model coefficients. The original ARDL data was converted into a time series, which is critical for the ARDL model's accuracy in capturing the temporal relationships among economic indicators, sentiment data, and Bitcoin prices. Since time series models like ARDL require stationarity, the original series was checked and, if necessary, adjusted to become stationary.

The process involved using an ARIMA model to forecast each of the relevant economic indicators (S&P 500, CPI, EUR/USD, GBP/USD, JPY/USD) and sentiment categories (Product_Technology, Investment, Community) over the next 30 days. By forecasting these predictors independently, the ARDL model could use these projections as input for the final Bitcoin price forecast. The coefficients obtained from fitting the ARDL model on the historical data were applied to the future values of the predictors. This approach allows the model to compute the predicted Bitcoin closing price based on the projected values of the economic indicators and sentiment categories. The equation follows the ARDL methodology by incorporating lagged values of predictors. See Figure 17 for the results.

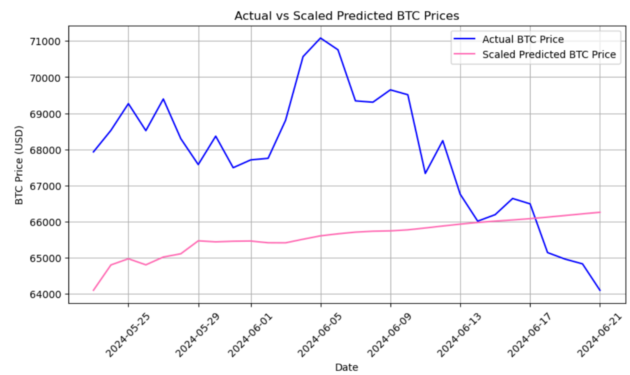
Figure 17.



Clearly, the ARDL model underestimated the Bitcoin prices. The predictions begin at approximately 57,347.09 and gradually increase, ending at 59,280.48. The model suggests moderate price growth with minimal volatility over the 30-day period. While the actual prices showed more volatility and a higher average range, the ARDL model predicted smoother trends, indicating that it might be better at capturing the overall direction but less effective in accounting for rapid changes and volatility in the market.

However, when applying the same logic as in the Linear Regression model and scaling the ARDL predictions, the results align more closely with the actual Bitcoin price movements. The scaled predictions start at approximately 64,096.20 and gradually increase to 66,257.13, as seen in Figure 18, reflecting a more realistic range when compared to the actual prices.

Figure 18.



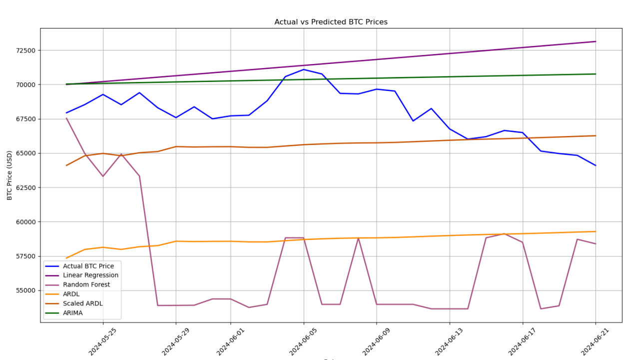
Although the scaling adjustment improves the overall prediction range and brings the model closer to the observed prices, the ARDL model still fails to capture the inherent volatility and sharper fluctuations evident in the actual data. This result underscores the ARDL model's tendency to predict smoother, more gradual trends, which can be beneficial for identifying long-term direction but less effective in markets like Bitcoin that are characterised by significant short-term volatility. Even with scaling, the ARDL model demonstrates its limitations in

forecasting sharp price spikes or dips, reinforcing the need for models that can better adapt to the erratic nature of cryptocurrency markets.

5. Comparison

Figure 18 illustrates the predicted BTC prices generated by four different models—Linear Regression, Random Forest, ARDL, and ARIMA—compared against the actual future BTC prices. Each model exhibits distinct performance characteristics when extrapolated beyond the original dataset.

Figure 18.



The ARIMA model, represented by the green line, shows a consistent linear upward trend, suggesting a steady growth prediction for Bitcoin prices. However, the ARIMA model fails to capture any of the significant fluctuations present in the actual data, leading to a generalised and unrealistic forecast, similar to the Linear Regression forecast.

The Linear Regression model, denoted by the purple line, also predicts a gradual increase in BTC prices but with a slightly sharper slope than ARIMA. While it maintains a clear upward trend, it too does not adequately account for the variability and volatility seen in the actual prices, potentially indicating oversimplification of the underlying patterns.

The Random Forest model, shown by the mauve line, offers more dynamic predictions, initially dropping sharply and then fluctuating considerably throughout the forecasted period. However, the model struggles with stability, leading to erratic and non-representative predictions, diverging significantly from actual trends.

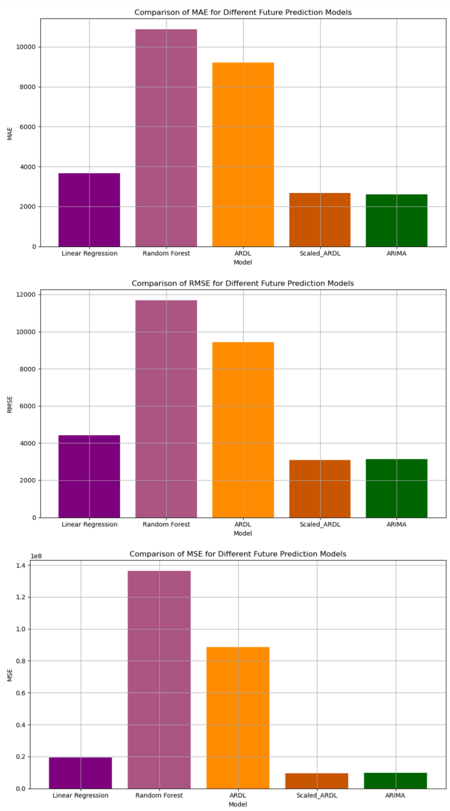
The ARDL model, represented by the orange line, stands out with a more stable and consistent prediction, closely following the broader trend, albeit missing the larger fluctuations. Although it does not fully capture the upward or downward movements, it maintains a level of prediction stability that is closer to the actual price movement when compared to the other models. When scaled, the ARDL model more closely reflects the Bitcoin prices, shown by the dark orange line.

When examining the models' predictive accuracy using the same metrics as before, the following is seen (Figure 19 and 20):

Figure 19.

Metric	Linear Regression	Random Forest	ARDL	Scaled ARDL	ARIMA
MAE	3662.730000	10867.280000	9202.010000	2678.040000	2596.010000
MSE	19436490.000000	136251600.000000	88746870.000000	9477730.000000	9913447.000000
RMSE	4408.680000	11672.690000	9420.560000	3078.590000	3148.560000

Figure 20.



The ARIMA model and the Scaled ARDL model stand out as the top-performing models in predicting Bitcoin prices outside of the dataset. The ARIMA model consistently produced low error rates, showcasing its strength in forecasting future values, particularly in capturing overall trends with minimal noise.

The Scaled ARDL model, after adjustment, significantly outperforms the original ARDL model. The scaling process effectively reduced the error rates, positioning this model as a competitive alternative to ARIMA for time-series forecasting. Linear Regression showed moderate performance but was less effective than ARIMA and Scaled ARDL in predicting future values. While it captured some key trends, it struggled with finer price fluctuations.

Interestingly, despite the excellent performance of the Random Forest model within the data, the Random Forest model performed poorly in this context, with the highest error metrics across the board. Although this model can be powerful in capturing nonlinear relationships, it failed to generalise effectively to unseen data in this scenario. This indicates that there was indeed significant overfitting occurring with the Random Forest model.

V. Chapter 5: Future Developments

The results and analysis presented in this study reveal several key areas for improvement and future research. The primary challenge identified is overfitting, which was most evident in the Random Forest and Linear Regression models. While these models performed exceptionally well within the training data, their predictions on unseen data highlighted their inability to generalise effectively. Addressing overfitting could involve implementing more sophisticated techniques such as cross-validation and regularisation, which would allow the models to strike a better balance between fit and generalizability.

Another area for future development involves exploring hybrid and ensemble models. By combining the strengths of ARIMA, ARDL, Random Forests, and Linear Regression, it may be possible to create a more robust predictive framework. For instance, leveraging ARIMA for capturing time-series trends while incorporating the non-linear predictive power of Random Forests could enhance both in-sample and out-of-sample performance.

The study also highlighted the significance of scaling adjustments, particularly for the ARDL model. Further research could explore more advanced scaling techniques or adaptive scaling methods that dynamically adjust based on market conditions. This could be especially valuable in volatile markets like Bitcoin, where rapid fluctuations are the norm.

Sentiment analysis remains a promising avenue for improving Bitcoin price predictions. While this study utilised free sentiment data from platforms like augmento.ai, refining the sentiment analysis process through more granular data or even employing natural language processing (NLP) techniques could lead to more accurate sentiment scores and better integration with economic indicators. Investigating additional sentiment sources or real-time sentiment tracking could further enhance the models' responsiveness to market shifts, though this may prove far more costly.

VI. Chapter 6: Conclusions

This research aimed to evaluate and compare various predictive models—ARIMA, ARDL, Random Forests, and Linear Regression—using freely accessible data sources to forecast Bitcoin

prices. The study focused on both in-sample and out-of-sample performance, aiming to identify which models are most effective in capturing Bitcoin's price dynamics.

The Random Forest model demonstrated strong predictive power within the training dataset, with very high R-squared value indicating a near-perfect fit. However, these results also pointed to significant overfitting, which compromised the models' effectiveness in forecasting future prices as seen. The Linear Regression model outperformed the ARDL model within the training data set, though it too showed signs of overfitting.

The ARIMA model and the Scaled ARDL model emerged as the most reliable when predicting prices outside the dataset, with the ARIMA model slightly outperforming the ARDL model. Despite the linear nature of this ARIMA model, it showed strong performance in capturing overall trends with minimal noise. The Scaled ARDL model, after adjustment, was competitive with ARIMA, demonstrating the value of scaling when dealing with non-linear market conditions.

The Random Forest model, despite its strong performance in the training data, struggled with erratic and non-representative predictions when applied to unseen data. This underlines the challenges of applying machine learning techniques in highly volatile financial markets where overfitting is a persistent issue.

The integration of sentiment analysis alongside traditional economic indicators offered valuable insights into Bitcoin price movements. However, further refinement is needed to fully leverage sentiment data, particularly in volatile environments where investor sentiment can shift rapidly.

While each model demonstrated strengths in specific contexts, no single model emerged as the definitive solution for predicting Bitcoin prices. The ARIMA and Scaled ARDL models provided the most balanced performance across in-sample and out-of-sample predictions, suggesting that a combination of econometric methods with proper adjustments might be the most effective approach. Moving forward, the integration of hybrid models, advanced scaling techniques, and more refined sentiment analysis presents a promising direction for future research. This study contributes to the broader discourse on financial forecasting, particularly in the realm of digital assets, by demonstrating the feasibility of using freely accessible data to build predictive models, while also highlighting the limitations that come with relying solely on such data sources.

VII. References

Altucher, J. (2024) How ETF's drive $100k #Bitcoin by April, YouTube. Available at: https://www.youtube.com/watch?v=CxFNyi-niwM&list=PLWTRLGkkf1kTTbIDdv-P6ykul7a8iUXYf&index=54 (Accessed: 28 April 2024).

Balcilar, M. et al. (2017) 'Can volume predict Bitcoin returns and volatility? A quantiles-based approach', Economic Modelling, 64, pp. 74–81. doi:10.1016/j.econmod.2017.03.019.

Basar, S. (2024) Blackrock Bitcoin ETF fastest to reach $10bn, Markets Media. Available at: https://www.marketsmedia.com/blackrock-bitcoin-etf-fastest-to-reach-10bn/#:~:text=BREAKING%3A%20BlackRock's%20Bitcoin%20ETF%2C%20%24,reached%20the%20%2410%20billion%20mark. (Accessed: 29 April 2024).

Benzinga (2024) Bitcoin spot etfs net $31.6m inflows as Hong Kong Readies for ETF launches by Benzinga, Investing.com UK. Available at: https://uk.investing.com/news/stock-market-news/bitcoin-spot-etfs-net-316m-inflows-as-hong-kong-readies-for-etf-launches-3451690 (Accessed: 28 April 2024).

Bitcoin.com (2024) 'How bitcoin transactions work: Learn all about BTC: Get started with Bitcoin.com' (2024) Bitcoin.com. Available at: https://www.bitcoin.com/get-started/how-bitcoin-transactions-work/ (Accessed: 24 March 2024).

Bhatt, S., Ghazanfar, M. and Amirhosseini, M.H. (2023) 'Machine learning based cryptocurrency price prediction using historical data and social media sentiment', *Machine Learning & Applications* [Preprint]. doi:10.5121/csit.2023.131001.

Bouoiyour, J. and Selmi, R. (2015) What Does Bitcoin Look Like? Annals of Economics and Finance, 16, pp.449-492.

Bouoiyour, J., Selmi, R., Tiwari, A.K. & Olayeni, O.R. (2016) What drives Bitcoin price? Economics Bulletin, 36(2), pp.843-850.

Bouri, E. et al. (2018) 'Testing for asymmetric nonlinear short- and long-run relationships between Bitcoin, aggregate commodity and gold prices', Resources Policy, 57, pp. 224–235. doi:10.1016/j.resourpol.2018.03.008.

Chen, J. (2023) 'Analysis of bitcoin price prediction using machine learning', *Journal of Risk and Financial Management*, 16(1), p. 51. doi:10.3390/jrfm16010051.

Choi, J.W. and Choi, Y.K. (2024) 'The prediction of bitcoin price through gold price using long short-term memory model', *IAES International Journal of Artificial Intelligence (IJ-AI)*, 13(1), p. 909. doi:10.11591/ijai.v13.i1.pp909-916.

Ciaian, P., Rajcaniova, M. and Kancs, d'Artis (2015) 'The economics of Bitcoin Price Formation', Applied Economics, 48(19), pp. 1799–1815. doi:10.1080/00036846.2015.1109038.

Coinage (2024) *Bloomberg analyst makes $97 billion prediction for Bitcoin etfs - thestreet crypto: Bitcoin and cryptocurrency news, advice, analysis and more*, *thestreet.com*. Available at: https://www.thestreet.com/crypto/markets/bloomberg-analyst-makes-97-billion-prediction-for-bitcoin-etfs (Accessed: 28 April 2024).

Critien, J.V., Gatt, A. and Ellul, J. (2022) 'Bitcoin price change and trend prediction through Twitter sentiment and Data Volume', Financial Innovation, 8(1). doi:10.1186/s40854-022-00352-7.

De Mott, F. (2024) *Bitcoin could soar 266% to $250,000 next year if ETF inflows Stay Strong, standard chartered says*, *Business Insider*. Available at:

https://www.businessinsider.com/bitcoin-price-outlook-250000-etf-inflows-fx-reserves-crypto-gold-2024-3 (Accessed: 28 April 2024).

Dimitriadou, A. and Gregoriou, A. (2023) 'Predicting bitcoin prices using machine learning', *Entropy*, 25(5), p. 777. doi:10.3390/e25050777.

Frohmann, M. *et al.* (2023) 'Predicting the price of bitcoin using sentiment-enriched time series forecasting', *Big Data and Cognitive Computing*, 7(3), p. 137. doi:10.3390/bdcc7030137.

Garcia, D. et al. (2014) 'The digital traces of bubbles: Feedback cycles between socio-economic signals in the Bitcoin economy', Journal of The Royal Society Interface, 11(99), p. 20140623. doi:10.1098/rsif.2014.0623.

GBTC (2024) *What is GBTC?*, *Grayscale Bitcoin Trust ETF (GBTC)*. Available at: https://etfs.grayscale.com/gbtc (Accessed: 28 April 2024).

Georgoula, I. et al. (2015) 'Using time-series and sentiment analysis to detect the determinants of bitcoin prices', SSRN Electronic Journal [Preprint]. doi:10.2139/ssrn.2607167.

Gurdgiev, C. and O'Loughlin, D. (2020) 'Herding and anchoring in cryptocurrency markets: Investor reaction to fear and uncertainty', SSRN Electronic Journal [Preprint]. doi:10.2139/ssrn.3517006.

Hao, M. *et al.* (2022) 'Bitcoin price prediction based on machine learning and Granger Causality Test', *Proceedings of the 2022 2nd International Conference on Economic Development and Business Culture (ICEDBC 2022)*, pp. 342–348. doi:10.2991/978-94-6463-036-7_51.

Hayes, A. (2015) 'Cryptocurrency value formation: An empirical analysis leading to a cost of production model for valuing Bitcoin.', SSRN Electronic Journal [Preprint]. doi:10.2139/ssrn.2648366.

Hodson, T.O. (2022) ‘Root-mean-square error (RMSE) or mean absolute error (mae): When to use them or not’, Geoscientific Model Development, 15(14), pp. 5481–5487. doi:10.5194/gmd-15-5481-2022.

Hunt, J. (2024) *Grayscale's GBTC is down 50% in bitcoin holdings since spot etfs launch*, *The Block*. Available at: https://www.theblock.co/post/288590/grayscale-gbtc-halving-bitcoin-holdings-since-spot-etfs-launch (Accessed: 28 April 2024).

Kang, R., Hwang, S. and Shin, J. (2024) ‘Cryptocurrency prices and news sentiment: Which exerts influence on the other?’, *SSRN Electronic Journal* [Preprint]. doi:10.2139/ssrn.4714113.

Kim, J.H. (2019) ‘Multicollinearity and misleading statistical results’, Korean Journal of Anesthesiology, 72(6), pp. 558–569. doi:10.4097/kja.19087.

Kjærland, F. et al. (2018) ‘An analysis of Bitcoin's price dynamics’, Journal of Risk and Financial Management, 11(4), p. 63. doi:10.3390/jrfm11040063.

Kristoufek, L. (2015) ‘What are the main drivers of the bitcoin price? evidence from wavelet coherence analysis’, PLOS ONE, 10(4). doi:10.1371/journal.pone.0123923.

Lu, J. *et al.* (2023) ‘Research on the price prediction of Bitcoin and gold based on Random Forest Model’, *Proceedings of the 2nd International Conference on Information, Control and Automation, ICICA 2022, December 2-4, 2022, Chongqing, China* [Preprint]. doi:10.4108/eai.2-12-2022.2327931.

Modi, P.D. *et al.* (2023) ‘A data-driven deep learning approach for Bitcoin Price forecasting’, *2023 24th International Conference on Digital Signal Processing (DSP)* [Preprint]. doi:10.1109/dsp58604.2023.10167930.

Mudassir, M. et al. (2020) ‘Time-series forecasting of bitcoin prices using high-dimensional features: A machine learning approach’, Neural Computing and Applications [Preprint]. doi:10.1007/s00521-020-05129-6.

Mukaka, M.M. (2012) Statistics corner: A guide to appropriate use of correlation coefficient in medical research, Malawi medical journal : the journal of Medical Association of Malawi. Available at: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3576830/#:~:text=For%20example%2C%20a%20correlation%20coefficient,use%20the%20most%20appropriate%20one. (Accessed: 19 August 2024).

Narayanan, A. et al. (2016) ‘Bitcoin and cryptocurrency technologies ’, Network Security, 2016(8), p. 4. doi:10.1016/s1353-4858(16)30074-5.

Pano, T. and Kashef, R. (2020) ‘A complete vader-based sentiment analysis of Bitcoin (BTC) tweets during the era of covid-19’, Big Data and Cognitive Computing, 4(4), p. 33. doi:10.3390/bdcc4040033.

Polasik, M. et al. (2015) ‘Price fluctuations and the use of bitcoin: An empirical inquiry’, International Journal of Electronic Commerce, 20(1), pp. 9–49. doi:10.1080/10864415.2016.1061413.

Reiff, N. (2024) *Bitcoin Futures ETF: Definition, how it works, and how to invest*, *Investopedia*. Available at: https://www.investopedia.com/investing/bitcoin-etfs-explained/ (Accessed: 28 April 2024).

Serafini, G. et al. (2020) ‘Sentiment-driven price prediction of the Bitcoin based on statistical and deep learning approaches’, 2020 International Joint Conference on Neural Networks (IJCNN) [Preprint]. doi:10.1109/ijcnn48605.2020.9206704.

Strack, B. (2024) *Successful Hong Kong crypto ETF flows would pale in comparison to US funds*, *Blockworks*. Available at: https://blockworks.co/news/hong-kong-crypto-etf-predicted-inflows (Accessed: 28 April 2024).

Tangem (2024) What is a spot bitcoin ETF? how does it work?, What is a Spot Bitcoin ETF? How Does It Work? | Tangem Blog. Available at: https://tangem.com/en/blog/post/spot-bitcoin-etf/ (Accessed: 01 August 2024).

Vardai, Z. (2024) *Hong Kong bitcoin and ether etfs officially approved to start trading on April 30*, *Cointelegraph*. Available at: https://cointelegraph.com/news/hong-kong-bitcoin-ether-etfs-start-trading-april-30 (Accessed: 28 April 2024).

Wołk, K. (2019) ‘Advanced social media sentiment analysis for short‐term cryptocurrency price prediction’, Expert Systems, 37(2). doi:10.1111/exsy.12493.

Young, M. (2023) *Gold reaches record high amid predictions of bitcoin market cap overtake*, *BeInCrypto*. Available at: https://beincrypto.com/gold-price-all-time-high-bitcoin-market-cap-flip/ (Accessed: 28 April 2024).

Zhen , S. and Xue, J. (2024) *Hong Kong set to approve its first spot bitcoin etfs in April, sources say | Reuters*, *reuters.com*. Available at: https://www.reuters.com/business/finance/hong-kong-set-approve-its-first-spot-bitcoin-etfs-april-sources-say-2024-04-10/ (Accessed: 28 April 2024).